Japanese Journal of Applied Physics

The Japan Society of Applied Physics

REGULAR PAPER

Direct growth of $NbSe_2$ and $TiSe_2$ thin flakes via deposited film selenization

To cite this article: Ren Hachiya *et al* 2020 *Jpn. J. Appl. Phys.* **59** SCCB25

View the article online for updates and enhancements.

This content was downloaded from IP address 109.94.222.113 on 21/03/2020 at 07:55

Japanese Journal of Applied Physics **59**, SCCB25 (2020) **REGULAR PAPER**
https://doi.org/10.7567/1347-4065/ab5913

Direct growth of $NbSe_2$ and $TiSe_2$ thin flakes via deposited film selenization

Ren Hachiya[1], Noriyuki Urakami[1,2*], and Yoshio Hashimoto[1,2]

[1]*Department of Electrical and Computer Engineering, Faculty of Engineering, Shinshu University, 4-17-1 Wakasato, Nagano 380-8533, Japan*
[2]*Research Initiative for Supra-Materials, Shinshu University, 4-17-1 Wakasato, Nagano 380-8533, Japan*

*E-mail: urakami@shinshu-u.ac.jp

Received July 9, 2019; revised November 6, 2019; accepted November 18, 2019; published online December 20, 2019

We report the direct growth of thin flakes of $NbSe_2$ and $TiSe_2$ on c-plane sapphire substrate via the selenization of deposited Nb and Ti films, respectively. The effect of the cooling rate on the shapes of the $NbSe_2$ thin flakes, for example, hexagons and triangles, was studied. In particular, rapid cooling after selenization of the deposited Nb film was found to be effective for growing triangular $NbSe_2$ flakes. Raman spectroscopy of one sample prepared by rapid cooling revealed an unidentified vibration peak in addition to the prominent characteristic peaks of 2H-$NbSe_2$. The Ti film could only be selenized by using sodium chloride as a transport agent, which resulted in the growth of $TiSe_2$ thin flakes having a half-hexagonal shape without the incorporation of residual materials. Temperature-dependent resistances measurement, including the measurements of the temperature of maximum resistance and R_{max}/R_{293K} ratio, indicated a conventional charge-density wave transition in the $TiSe_2$ thin flakes.
© 2019 The Japan Society of Applied Physics

Supplementary material for this article is available online

1. Introduction

Graphene has received considerable attention as a two-dimensional optoelectronic material with high performance, low cost, and flexible applications. Its high electron mobility is exploited in electrical device applications such as logic circuits. However, its low on/off ratio limits the usefulness of graphene channel transistors.[1–5] Layered transition metal dichalcogenides (TMDCs) materials, including molybdenum sulfide (MoS_2) and tungsten sulfide (WS_2), have been studied recently as alternatives to graphene.[6–8] TMDCs can be easily cleaved and the number of layers can be atomically controlled because of the van der Waals force between the stacked layers. Therefore, the dependence of various physical properties on the layer number has drawn attention to these next-generation solid state materials. In addition, it is expected that devices based on novel principles can be obtained by realizing of various electronic states by selecting appropriate combinations of transition metals, chalcogen elements, and crystal structures.

The insulator-metal transition (IMT) originates in electron localization and delocalization due to electron–electron and electron–phonon interactions and leads to nontrivial (anomalous) changes in the electronic properties. Therefore, the IMT has been recognized as one of the most important phenomena for the realization of high-performance switching, steep switching slopes, and novel functional devices beyond complementary metal-oxide-semiconductor electronics. The IMT has been observed in TMDCs systems via charge-density waves (CDWs), which are due to strong lattice coupling. The phase transitions of these CDWs, which can be electrically controlled, result in changes in the conductivity due to coupling and decoupling from the lattice. Strongly correlated electronic phenomena have been studied in the TMDCs $NbSe_2$ and $TiSe_2$; for example, the electron–hole coupling has been enhanced by decreasing the sample dimensions.

TMDCs films and flakes have generally been prepared by mechanical exfoliation from bulk single crystals grown by chemical vapor transport (CVT). CVT usually requires long growth times on the order of at least several days. Additionally, these exfoliated films and flakes are small and have random thickness distributions. Thus, the thermal treatment of deposited transition metal films under ambient chalcogen vapor, which can be used for direct growth of TMDCs on various substrates, has been investigated. In this so-called vapor–solid growth method,[9,10] the chalcogen vapor reacts with the transition metal. By using this method, TMDCs such as MoS_2, WS_2 and ReS_2[11,12] have been synthesized easily with high quality in relatively short times (from several tens of minutes to a few hours). Therefore, this method is a promising approach to growing TMDCs crystals. However, there are few reports on vapor–solid growth of certain TMDCs crystals, including $NbSe_2$ and $TiSe_2$. Metallic Ti is easily oxidized and thus difficult to selenize owing the high decomposition temperature of oxidized Ti (above 1800 °C). The growth temperature of TMDCs is typically 1200 °C at most. Thus, experimental investigation of the growth process is important because Ti-based TMDCs have not yet been produced by vapor–solid growth. In this study, $NbSe_2$ and $TiSe_2$ are prepared by vapor–solid growth, and their electrical characteristics are investigated. The remainder of this paper is organized as follows: in t Sect. 2, the experimental methods (vapor–solid growth and film characterization) are summarized. Simple selenization of $NbSe_2$, and the effectiveness of using sodium chloride (NaCl) to promote selenization of Ti are described in Sect. 3. The results are summarized in Sect. 4.

2. Experimental procedure

All samples were fabricated by selenization of deposited metal films. First, epi-ready c-plane sapphire substrates (K-and-R Creation Co., Ltd., $\sim 10 \times 10\,mm^2$) were cleaned with acetone, methanol, and ultrapure water by ultrasonic washing. Next, Nb or Ti films were deposited on the substrates by electron-beam evaporation (ANELVA 980–7104) using metallic Nb sponges (3 N, Kojundo Chemical Lab. Co. Ltd.) and Ti sponges (4 N, Kojundo Chemical Lab. Co. Ltd.) as the sources. The chamber pressure was $\sim 1 \times 10^{-6}$ Pa during deposition. The thickness of the deposited metals was approximately 120 nm. Then, the Nb-deposited sapphire substrates and Se powder (6 N, Yamanaka

 © 2019 The Japan Society of Applied Physics

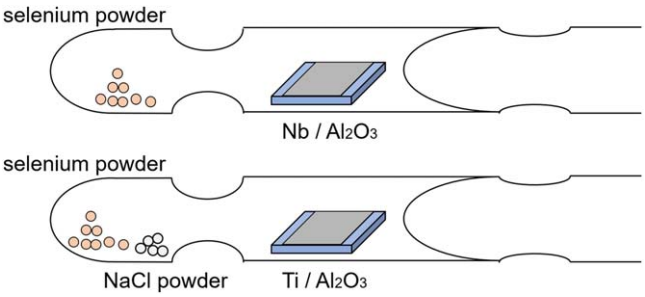

Fig. 1. (Color online) Schematic of the experimental setup of vapor solid growth. The Nb and Ti film deposited on the c-plane sapphire substrate is selenized in a sealed quartz tube.

Ceradyne, Inc.) were placed in a quartz tube for simple vapor–solid growth of $NbSe_2$. For $TiSe_2$ growth, the Ti-deposited sapphire substrates, Se powder and NaCl powder (3 N, Kojundo Chemical Lab. Co. Ltd.) were placed in a quartz tube. An improved halide-assisted growth method was used for $TiSe_2$ because of the increased volatility and reactivity of Ti films with native oxides. NaCl was used as the halide source and played a role similar to that of the transport agent in CVT.[13–15] The quartz tubes were pumped down to 3×10^{-3} Pa from the open ends using a rotary pump and a diffusion pump, after which they were sealed by high-temperature welding as shown in Fig. 1. The sealed quartz tubes were placed inside an electrical heating furnace and heated to a temperature between 700 °C and 1000 °C at a rate of 3.33 °C min^{-1}. After the reaction, the quartz tubes were cooled by two different processes. The first process consists of cooling to 500 °C at 1.67 °C min^{-1}, followed by natural cooling at room temperature. The second process is rapid cooling from the reaction temperature to room temperature immediately after heating. For $NbSe_2$ growth, both processes were used to investigate their effect on the shape. For $TiSe_2$ growth, only the first process was used. The Se pressures in the sealed quartz tubes was 1, 2, or 4 atm as determined from the vapor pressure considering the tube volume and amount of Se powder. The selenization time was fixed at 10 min. To measure the resistance of the flakes, two-terminal devices with Ti electrodes were fabricated on the flakes by ultraviolet photolithography and a lift-off process.

Microscopic Raman spectroscopy was performed to characterize the $NbSe_2$ and $TiSe_2$ thin flakes by analyzing the characteristic peaks attributed to in-plane and out-of-plane vibrational modes.[16–25] The 532 nm line of a diode-pumped solid-state laser was used for optical excitation. The Raman signal was dispersed in an 800 mm monochromator and a cooled charge-coupled device. The shapes of the $NbSe_2$ and $TiSe_2$ thin flakes were observed by scanning electron microscopy (SEM) and optical microscopy. Elemental analysis of the $NbSe_2$ and $TiSe_2$ thin flakes was performed using energy dispersive X-ray spectrometry (EDS). Electrical transport measurements were conducted using a prober at temperatures between 77 and 293 K, where the temperature was controlled by a heater and liquid nitrogen.

3. Results and discussions

We first report the effects of the growth temperature and cooling rate on the growth of $NbSe_2$ thin flakes. We then investigate the selenization of the Ti films using NaCl as a transport agent and discuss the reaction model. We finally present the electrical transport properties of $TiSe_2$ thin flakes measured at various temperatures.

3.1. Growth of $NbSe_2$ thin flakes by simple selenization of deposited Nb film

Figure 2(a) shows a typical optical image of $NbSe_2$ thin flakes grown at 700 °C and cooled at a controlled rate. $NbSe_2$ thin flakes approximately 10 μm in length were formed at high density after growth, as shown in Fig. S1, available online at stacks.iop.org/JJAP/59/SCCB25/mmedia in the supplementary material. The average thickness was estimated to be 2.1 μm. Hexagonal crystal structure was clearly observed. According to the SEM and EDS images of Nb and Se shown in Fig. S2 in the supplementary material, no impurities were incorporated into the $NbSe_2$ thin flakes. This result demonstrates that $NbSe_2$ thin flakes were formed directly on sapphire substrates, indicating that thermal decomposition of the native oxide of the Nb film at the processing temperature, selenization of the Nb film, and growth of $NbSe_2$ thin flakes occurred. Next, the cooling process after selenization of the Nb film was investigated. Figure 2(b) shows an optical image of $NbSe_2$ thin flakes grown at 700 °C and cooled rapidly after heating and selenization. Triangular flakes, which are often seen in TMDCs growth, are also observed in addition to the hexagonal flakes. The average thickness was estimated to be 880 nm. This result indicates that triangular $NbSe_2$ thin flakes were initially formed at the growth temperature. The flakes became hexagonal during controlled cooling because the hexagonal structure is the more energetically stable.[26,27] The shape of the $NbSe_2$ thin flakes can thus be controlled by

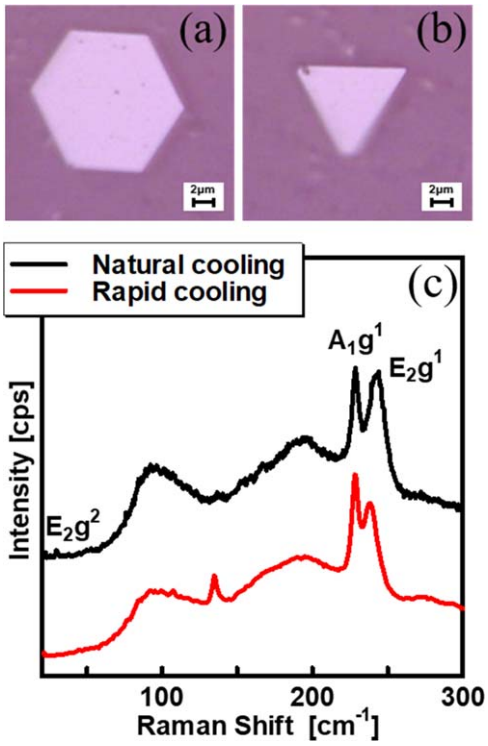

Fig. 2. (Color online) Optical microscopy images of the $NbSe_2$ thin flakes on the c-plane sapphire substrate, which were prepared using (a) controlled slow and (b) rapid cooling processes, respectively. (c) Raman spectra collected from $NbSe_2$ thin flakes. Upper trace: controlled slow cooling process, lower trace: rapid cooling process.

© 2019 The Japan Society of Applied Physics

selecting the appropriate cooling process after selenization of Nb films.

Figure 2(c) shows the Raman spectra of $NbSe_2$ thin flakes obtained by each cooling processes (controlled slow cooling and rapid cooling). The prominent characteristic peaks at approximately 227 and 237 cm^{-1}, which correspond to the out-of-plane A_{1g} and in-plane E^1_{2g} vibrational modes of 2H-$NbSe_2$, respectively, were detected in both samples, indicating the formation of 2H-$NbSe_2$.[16,19] Although the peak positions in the Raman spectra indicate that both thin flakes should be identified as the 2 H polytype, the characteristic peak of the in-plane E^2_{2g} vibrational mode at 28.5 cm^{-1} was detected only for the flake treated by controlled slow cooling. The E^2_{2g} vibration mode has also been observed as the first-order active mode of 2H-$NbSe_2$ in addition to the out-of-plane A_{1g} and in-plane E^1_{2g} modes described above.[28] In contrast, a strong peak of unknown origin at 135 cm^{-1} was detected for the rapidly cooled sample. It is possible that the peak originated from an anomalous E_{1g} vibration mode, which has been predicted from the linear chain model.[18] However, the unknown peak observed in our study is not strictly consistent with the predicted vibration mode because the peak position of the E_{1g} vibration mode is 133 cm^{-1}. Another possibility is the formation of other polytypes such as 1 T and 3 R in the 2 H polytype crystal, which are ovserved for NbS_2 and have not yet been detected for $NbSe_2$. A more detailed investigation including the use of scanning tunneling microscopy to confirm the lattice order is needed.

3.2. Growth of $TiSe_2$ thin flakes by improved selenization of deposited Ti film

Figure 3 shows the Raman spectra of Ti films selenized with and without NaCl as a transport agent at a growth temperature of 800 °C and a selenization pressure of 1 atm. When NaCl was not used, the characteristic peaks were located at approximately 144 and 240 cm^{-1}, indicating rutile type TiO_2. The peaks reveal that selenization could be incomplete, because the decomposition temperature of the Ti oxide is significantly higher than the selenization temperature. In contrast, when NaCl was used, the characteristic peaks were located at approximately 136 and 200 cm^{-1}, indicating 1T-$TiSe_2$. The Ti film was sufficiently selenized by using NaCl as a transport agent. Raman signals could be detected from $TiSe_2$ thin flakes with NaCl/Se ratios of 6.25%–25%. The inset of Fig. 3 shows an optical image of a typical 1T-$TiSe_2$ thin flake grown using NaCl. The average thickness was estimated to be 990 nm. The flake has a half-hexagonal shape. This shape was also observed in other samples prepared under different conditions, as shown in Fig. S3 in the supplementary material. The half-hexagonal shape results from direct growth along the step edges of the c-plane sapphire substrate,[23] as reported previously in thin films of other TMDCs.[29] The total area of the thin flakes was much smaller than that of the precursor. Therefore, most of Ti film was desorbed by using NaCl during heating in ambient selenium vapor. Note, however, that the desorbed Ti should have contributed to the growth of $TiSe_2$ thin flakes because the average thickness was much thicker than that of the precursor Ti film. After the $TiSe_2$ thin flakes formed, the ratio of the average thickness multiplied by the covered area was estimated to be at most 6.5%. In addition, it is assumed that the broad Raman signal detected as the NaCl/Se increased to 50% is attributable to TiO_2, indicating the absence of $TiSe_2$ thin flakes. The absence of thin flakes at high NaCl contents is due to an excessive increase in the volatility of not only the oxidized Ti film, but also the selenized Ti film. The increased volatility of the latter results in re-formation of TiO_x after cooling.

The growth model of $TiSe_2$ thin flakes by selenization of a Ti film using NaCl as a transport agent is illustrated in Fig. 4(a). The Ti film could not be selenized in the absence of NaCl, indicating that the Ti film with native oxides is thermally stable at the synthesis temperature. Therefore, it is necessary to etch the surface of the oxidized Ti film. During heating in ambient selenium vapor with NaCl

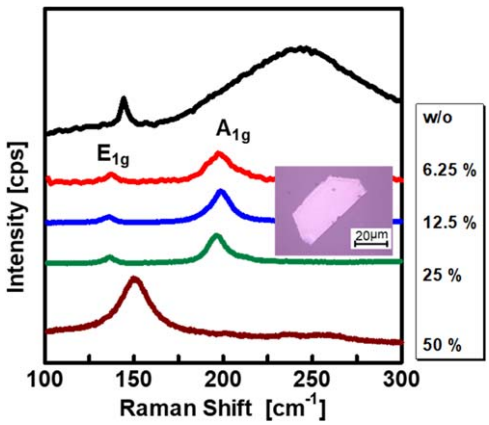

Fig. 3. (Color online) (a) Raman spectra collected from selenized Ti film with and without NaCl agent. $TiSe_2$ thin flakes depending on the NaCl/Se ratios in the range of 6.25%–50%.

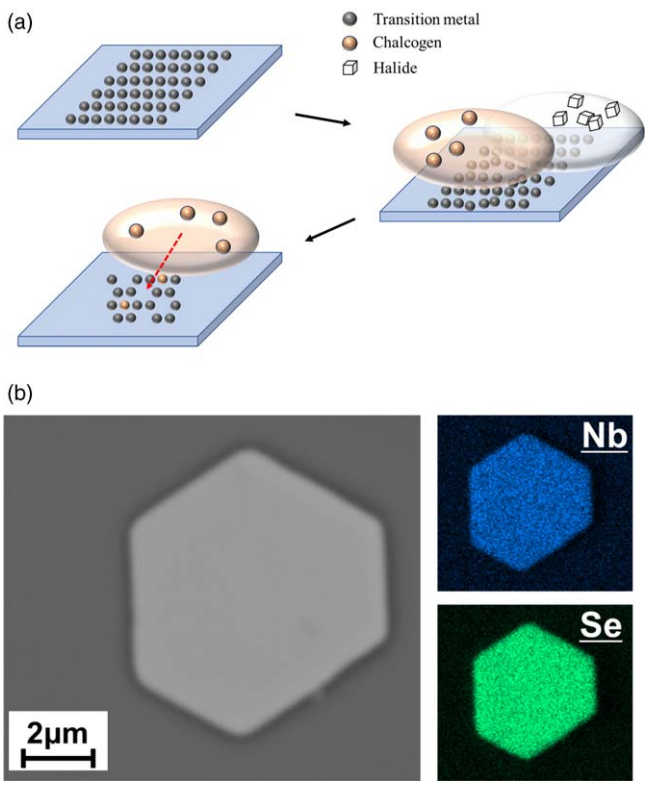

Fig. 4. (Color online) (a) Growth and reaction model for $TiSe_2$ thin flakes via the selenization of Ti film using NaCl agent. (b) SEM image of $TiSe_2$ thin flakes with half-hexagonal shape and EDS mapping of Ti and Se signals.

© 2019 The Japan Society of Applied Physics

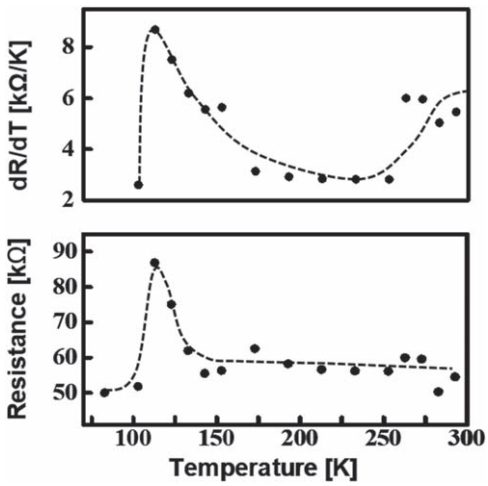

Fig. 5. Temperature dependence of resistance and temperature derivative of resistance for of the $TiSe_2$ thin flakes.

powder, chlorine (Cl_2) produced from NaCl reacted with the Ti film, etching its surface and increasing in the volatility. Selenization and growth of the Ti film were then realized, resulting in the formation of $TiSe_2$ thin flakes, where the role of NaCl is similar to that of a transport agent in CVT growth. Compared to CVT growth, selenization of the Ti film for the synthesis of $TiSe_2$ has the advantage of forming the film and flake-shaped $TiSe_2$ directly. Iodine (I_2) is typically used as the transport agent in CVT growth. However, its reactivity is too high compared to that of Cl_2. Therefore, because it is comparatively difficult to control the TMDCs film thickness, Cl_2 has also been considered as a transport agent in the growth of TMDCs thin films.[23] The absence of Na and Cl in the $TiSe_2$ thin flakes was confirmed by SEM imaging and EDS mapping, as shown in Fig. 4(b). It has been noted that alkali metals can intercalate into layered materials, causing changes in their physical properties. Here, we demonstrate that NaCl is a convenient transport agent for the growth of $TiSe_2$ thin flakes without the incorporation of residual materials.

To characterize the electronic transport properties of the $TiSe_2$ thin flakes, two Ti-electrodes were formed by a lift-off process. The resistance was measured at various temperatures ranging from 77 to 293 K. Figure 5 shows the temperature dependence of the resistance and the temperature derivative of the resistance of the $TiSe_2$ thin flakes. To highlight the features of our $TiSe_2$ sample, we focus on the maximum resistance and the characteristic temperature indicating the CDW transition. The maximum resistance occurred at approximately 120 K. The R_{max}/R_{293K} ratio, which has been used to characterize film quality,[30,31] was estimated to be 1.6. The value is comparable to or slightly higher than that of a $TiSe_2$ thin film grown on a sapphire substrate by molecular beam epitaxy.[32] Additionally, the characteristic temperature from the temperature derivative of the resistance was found to be approximately 170 K. This value agrees with a previous report.[23] The CDW transition temperature generally depends on the film quality because point defects associated with Se vacancies result in electron doping. These results suggest that high quality $TiSe_2$ crystals can be obtained by our growth method within a short time.

4. Summary

Thin flakes of $NbSe_2$ and $TiSe_2$ were grown directly on c-plane sapphire substrate by selenization of deposited Nb and Ti films, respectively. Hexagonal and triangular $NbSe_2$ thin flakes were formed using controlled slow cooling and rapid cooling processes, respectively. In addition to the prominent characteristic peaks such as the out-of-plane A_{1g} and in-plane E^1_{2g} vibrational modes of 2H-$NbSe_2$, Raman spectroscopy revealed the in-plane E^2_{2g} vibrational mode in the samples that were cooled slowly and an undetermined vibration in the samples that were cooled rapidly. The origin of the latter might be an anomalous E_{1g} vibration mode or the formation of other polytypes such as 1 T and 3 R in addition to 2 H. Half-hexagonal $TiSe_2$ thin flakes were grown successfully only when NaCl was used as a transport agent. When excessive NaCl was used, $TiSe_2$ thin flakes did not grow. We confirmed that no Na and Cl incorporated into the $TiSe_2$ thin flakes. The temperature dependence of the resistance of the $TiSe_2$ thin flakes indicated the CDW transition. Additionally, the temperature of maximum resistance and R_{max}/R_{293K} ratio were consistent with previous reports, suggesting that NaCl is a convenient transport agent for growing high quality $TiSe_2$ thin flakes.

Acknowledgments

A part of this study was supported by the NIMS Nanofabrication Platform in the Nanotechnology Platform Project sponsored by the Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan. A part of this work was conducted using the analysis facilities at Global Aqua Innovation Center at Shinshu University (AICS).

1) G. H. Lee et al., ACS Nano **7**, 7931 (2013).
2) Q. H. Wang, K. K. Zadeh, A. Kis, J. N. Coleman, and M. S. Strano, Nat. Nanotechnol. **7**, 699 (2012).
3) K. S. Novoselov, Rev. Mod. Phys. **83**, 837 (2011).
4) A. K. Geim and I. V. Grigorieva, Nature **499**, 419 (2013).
5) K. J. Koski and Y. Cui, ACS Nano **7**, 3739 (2013).
6) B. Liu, L. Chen, G. Liu, A. N. Abbas, M. Fathi, and C. Zhou, ACS Nano **8**, 5304 (2014).
7) B. Singh, C. H. Hsu, W. F. Tsai, V. M. Pereira, and H. Lin, Phys. Rev. B **95**, 245136 (2017).
8) International Roadmap for Devise and System, "White paper on Beyond CMOS (IEEE, Piscataway, NJ, 2017).
9) C. Yim et al., Sci. Rep. **4**, 5458 (2014).
10) S. Wu, C. Huang, G. Aivazian, J. S. Ross, D. H. Cobden, and X. Xu, ACS Nano **7**, 2768 (2013).
11) L. Ma, D. N. Nath, E. W. Lee II, C. H. Lee, M. Yu, A. Arehart, S. Rajan, and Y. Wu, Appl. Phys. Lett. **105**, 072105 (2014).
12) N. Urakami, T. Okuda, and Y. Hashimoto, Jpn. J. Appl. Phys. **57**, 02CB07 (2018).
13) J. Zhou et al., Nature **556**, 355 (2018).
14) Z. Wang et al., Nanotechnology **28**, 325602 (2017).
15) S. Li, S. Wang, D.-M. Tang, W. Zhao, H. Xu, L. Chu, Y. Bando, D. Golberg, and G. Eda, Appl. Mater. Today **1**, 60 (2015).
16) W. G. Mcmullan and J. C. Irwin, Solid State Commun. **45**, 557 (1983).
17) X. Xi, L. Zhao, Z. Wang, H. Berger, L. Forró, J. Shan, and K. F. Mak, Nat. Nanotechnol. **10**, 765 (2015).
18) C. M. Pereira and W. Y. Liang, J. Phys. C: Solid State Phys. **15**, L991 (1982).
19) C. S. Wang and J. M. Chen, Solid State Commun. **14**, 1145 (1974).
20) A. Wypych, I. Bobowska, M. Tracz, A. Opasinska, S. Kadlubowski, A. K. Kaliszewska, J. Grobelny, and P. Wojciechowski, J. Nanomater. **2014**, 1 (2014).

 © 2019 The Japan Society of Applied Physics

21) F. Cheng, Z. Ding, H. Xu, S. J. R. Tan, I. Abdelwahab, J. Su, P. Zhou, J. Martin, and K. P. Loh, Adv. Mater. Interfaces **5**, 1800429 (2018).
22) L. A. A. Rodríguez, M. Pianassola, and D. N. Travessa, Mater. Res. **20**, 96 (2017).
23) J. Wang, H. Zheng, G. Xu, L. Sun, D. Hu, Z. Lu, L. Liu, J. Zheng, C. Tao, and L. Jiao, J. Am. Chem. Soc. **138**, 16216 (2016).
24) D. L. Duong, G. Ryu, A. Hoyer, C. Lin, M. Burghard, and K. Kern, ACS Nano **11**, 1034 (2017).
25) X. Zhang, Q. H. Tan, J. B. Wu, W. Shi, and P. H. Tan, Nanoscale **8**, 6435 (2016).
26) S. Wang, Y. Rong, Y. Fan, M. Pacios, H. Bhaskaran, K. He, and J. H. Warner, Chem. Mater. **26**, 6371 (2014).
27) L. Yang, P. Liu, J. Li, and B. Xiang, Catalystes **7**, 285 (2017).
28) A. Mialitsin, J. Phys. Chem. Solids **72**, 568 (2011).
29) L. Chen, B. Liu, M. Ge, Y. Ma, A. N. Abbas, and C. Zhou, ACS Nano **9**, 8368 (2015).
30) Y. Wang, M. Nakano, Y. Kashiwabara, H. Matsuoka, and Y. Iwasa, Appl. Phys. Lett. **113**, 073101 (2018).
31) D. J. Campbell et al., Phys. Rev. Mater. **3**, 053402 (2019).
32) F. J. Di Salvo, D. E. Moncton, and J. V. Waszczak, Phys. Rev. B **14**, 4321 (1976).

 © 2019 The Japan Society of Applied Physics